High-efficiency, Foundry-compatible Silicon Grating Couplers Using Focusing and Apodized Designs

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Abstract: Efficient fibre-to-chip coupling remains a primary bottleneck in silicon photonics. Standard two-dimensional (2D) grating optimizations often fail when translated to three-dimensional (3D) layouts, suffering from severe spectral degradation due to non-ideal diffraction and parasitic Fabry-Perot interference. In this simulation study, we propose a single-etch 3D grating coupler design that systematically resolves these problems. By combining a focusing geometry with an apodized grating profile, we reshape the coupling strength and redirect the diffracted wavefront to suppress multi-path reflections. Designed for a standard 220-nm silicon-on-insulator platform without complex reflectors, the device achieves a simulated peak transmission of 0.555 (-2.56 dB) perfectly locked at 1550 nm, alongside a broad 61.05-nm bandwidth. The resulting spectrum is smooth and oscillation-free. This work provides a possibly straightforward, tape-out-ready methodology for realizing high-efficiency, spectrally stable optical interfaces.

1. Introduction

Silicon photonics provides a compact and CMOS-compatible platform for large-scale photonic integration, but efficient fibre-to-chip coupling remains a central challenge because the mode field diameter of a standard single-mode fibre at 1550 nm is much larger than the cross-section of a silicon waveguide. Among the available interface schemes, grating couplers are especially attractive because they allow out-of-

plane coupling, flexible placement on chip, and wafer-scale optical testing. These advantages make them one of the most widely used optical I/O structures in silicon photonic circuits, despite their stronger wavelength sensitivity and generally lower efficiency than edge couplers.[1]

A grating coupler operates by converting an in-plane guided mode into an out-of-plane radiated field through phase matching between the waveguide propagation constant and the grating momentum. Its central design relation can be written as Equation (1.1)

$$\beta - k_x = m\frac{2\pi}{\Lambda} \tag{1.1}$$

The expression show that the coupling wavelength and radiation angle are jointly determined by the grating period, effective index and fibre tilt angle. In practice, however, phase matching alone is not sufficient to guarantee high coupling efficiency, because the final performance is also limited by substrate leakage, back-reflection and mode mismatch between the diffracted field and the fibre mode.

Figure 1 shows the schematic cross-section of a shallow-etched silicon grating coupler and defines the key geometric and optical parameters used throughout this work. The structure consists of a silicon waveguide core, a top cladding, a buried oxide layer and a silicon substrate. The main design variables include the grating period Λ, tooth width W, fill factor $ff = W/\Lambda$, and etch depth e_d, while the optical response is characterized by the upward-radiated power P_{up}, downward leakage P_{down}, substrate loss P_{sub}, reflected power P_r, and guided wave power P_{wg}. The diffraction angle can be defined in the cladding, in air, or in the fibre, depending on the coupling configuration.

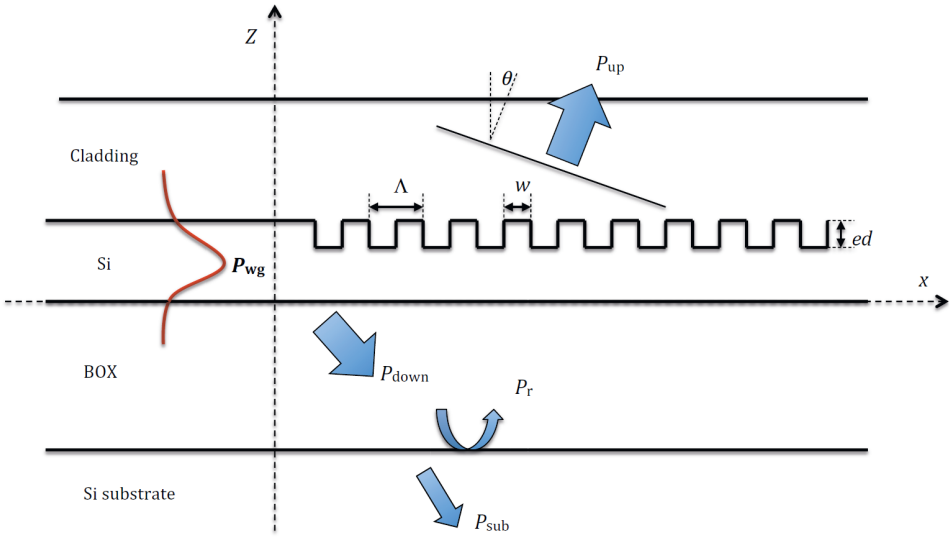

Figure 1. Schematic cross-section diagram of a grating coupler.[1]

For this reason, modern grating-coupler design relies on numerical optimization in addition to analytical estimates. Two-dimensional FDTD is commonly used for rapid initial design because it captures the main longitudinal coupling physics with much lower computational cost, while three-dimensional FDTD is required to verify the final device and account for full vectorial and lateral effects. Prior studies also show that perfectly vertical or near-vertical grating operation can suffer from strong back-reflection, which may induce Fabry-Perot oscillations and distort the transmission spectrum. A small fibre tilt, as well as structural redesign of the grating itself, is therefore often necessary to suppress such parasitic interference.

To improve coupling efficiency without introducing complex non-standard processes (e.g., bottom metal reflectors or poly-silicon overlays), wavefront engineering must be integrated into the physical layout. In this work, we demonstrate a highly robust, single-etch 3D grating coupler on a standard 220-nm silicon-on-insulator (SOI) platform. By implementing a focusing geometry and an apodized grating profile, we systematically suppress multi-path parasitic reflections and optimize the spatial overlap with the optical fiber mode. The resulting device elegantly recovers the ideal 2D spectral envelope in a fully 3D realistic environment, offering a straightforward, tape-out-ready methodology for high-efficiency optical interfaces.

2. Device Design and Simulation Methodology

2.1. Numerical Simulation Setup

Full-wave electromagnetic simulations of the proposed grating couplers were conducted using both two-dimensional (2D) and three-dimensional (3D) finite-difference time-domain (FDTD) solvers implemented in the software Ansys Lumerical FDTD. Silicon (Si) and silicon dioxide (SiO_2) material properties were described by the Palik dispersion models, which provide accurate refractive-index data across the target 1550 nm telecommunication band. All peripheral boundaries of the computational domain were terminated with perfectly matched layers (PML) to absorb outgoing radiation and eliminate artificial reflections.

A non-uniform conformal meshing scheme was adopted to balance accuracy and computational cost. In the grating coupling region, where the shallow 80 nm etch depth defines the critical sub-wavelength features, localized mesh override regions were applied with refined cell sizes of $\Delta x, \Delta y, \Delta z \leq 10\text{-}20$ nm. A TE-polarized Gaussian beam source (waist radius 5.2 μm, incidence angle 20°) was launched to emulate fiber excitation. Coupling efficiency into the fundamental TE mode of the 500 nm single-mode silicon waveguide was extracted directly from mode-expansion monitors placed at the waveguide output port, with supporting power transmission monitors used to verify field profiles and energy conservation. These settings ensured high-fidelity, reproducible results for both the preliminary 2D uniform-grating studies and the final 3D focusing, apodized designs.

2.2. 2D uniform straight grating model and Phase-Matching Mechanics

The device architecture is based on a standard foundry-compatible SOI platform, comprising a 220-nm silicon device layer, a 2 μm buried oxide (BOX) layer, and an oxide top cladding. To establish a robust design, the longitudinal coupling mechanics were first isolated using a 2D model. The initial grating period (Λ) was determined by Equation (2.1.1) using the Bragg phase-matching condition.

$$n_{\text{eff}} - n_c \sin\theta = \frac{\lambda}{\Lambda} \tag{2.1}$$

where n_{eff} is the effective index in the grating region, n_{c} is the cladding refractive index, θ is the radiation angle and Λ is the grating period. For a shallow-etched uniform grating, the grating-region effective index was approximated by

$$n_{\text{eff}} = ff\, n_{\text{eff},1} + (1 - ff)\, n_{\text{eff},2} \tag{2.2}$$

with ff the fill factor and $n_{\text{eff},1}$, $n_{\text{eff},2}$ the effective indices of the unetched and etched regions, respectively. This analytic model is not intended to replace full-wave simulation, but it provides a physically transparent and fabrication-aware starting point for subsequent optimization.

2.3. 3D focusing-apodized grating design

Directly extending the 2D model into a 3D straight-grating layout introduces effects that are not present in 2D. The finite width of the structure causes lateral field spreading and causes multi-path interference. As a result, the coupling spectrum is no longer set only by the designed grating response, but is also distorted by parasitic interference.

To address this limitation, the straight grating was redesigned as a 3D focusing-apodized grating coupler. The redesign therefore targeted both the radiated wavefront and the local coupling strength. To direct the diffracted field into a submicron waveguide, the grating trenches were mapped onto confocal ellipses defined by Equation (2.3).

$$m\lambda = n_{\text{eff}}\sqrt{x^2 + y^2} - x n_0 \sin\theta \tag{2.3}$$

This focusing geometry redirects the radiated field toward a common focal region and reduces the need for a long linear taper, as shown schematically in **Figure 2**.

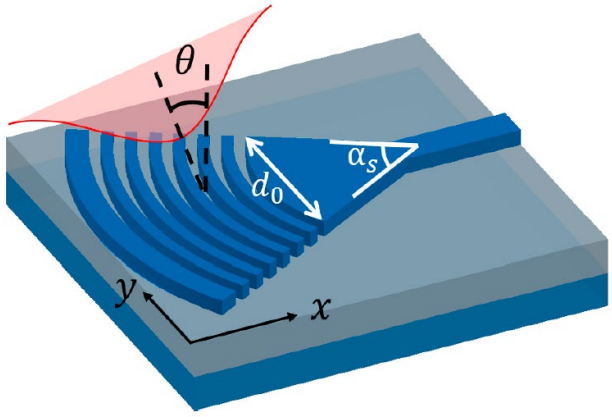

Figure 2. Schematic diagram of a focusing GC with curved grating lines.[2]

The final structure consisted of 16 curved grating teeth, with the period chirped from 0.635 to 0.675 μm and the fill factor chirped from 0.72 to 0.41. The etch depth was increased to 80 nm, and the focused field was coupled into a 500 nm-wide single-mode silicon waveguide. A TE-polarized Gaussian source with 20° incidence and 5.2 μm waist radius was used throughout, so that the focusing-apodized design could be compared directly with the straight-grating cases.

In this structure, the two design elements serve different functions. The curved grating lines redirect the radiated field toward the taper entrance. The chirped period and fill factor smooth the spatial distribution of scattering along the grating, reduce abrupt radiation, and improve overlap with the incident Gaussian mode.

3. Results

3.1. 2D optimized straight grating coupler

The optimized 2D straight grating exhibits a peak transmission of 0.692 at 1550 nm, as shown in **Figure 3**a), corresponding to a coupling loss of approximately -1.60 dB. The extracted 3 dB bandwidth is 54.19 nm. The spectrum is smooth and follows a near-ideal sinc^2 envelope, indicating that the longitudinal scattering strength is reasonably well balanced across the grating. In other words, the 2D optimization successfully aligns the phase-matching condition with the target wavelength while maintaining a

regular spectral profile. This result is consistent with the role of 2D FDTD as an efficient optimizer for shallow-etched TE grating couplers.

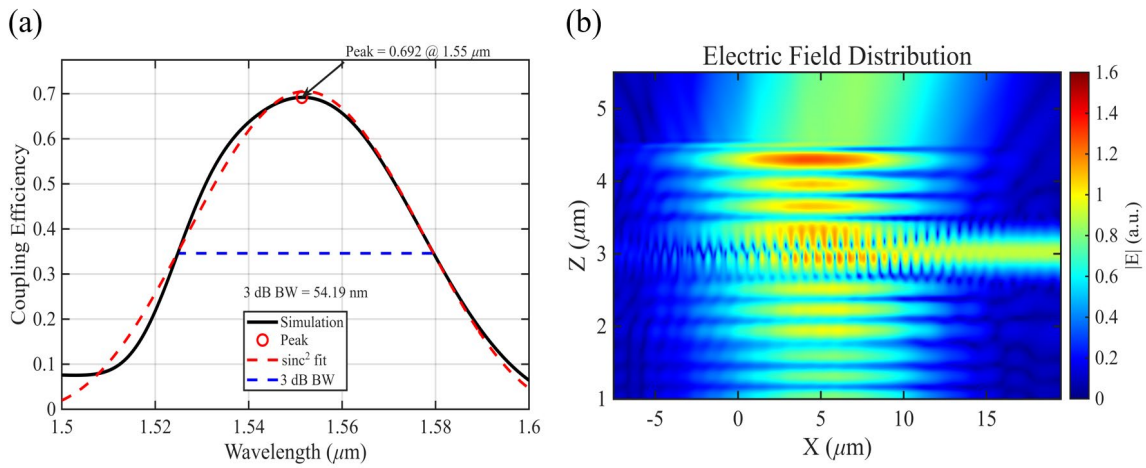

Figure 3. 2D optimized straight grating coupler. **a**) Simulated transmission spectrum of the optimized 2D uniform straight grating, showing a peak transmission of 0.692 at 1550 nm and a 3 dB bandwidth of 54.19 nm. **b**) Electric-field distribution of the same device, showing efficient diffraction and smooth coupling into the guided mode to the right.

The electric-field distribution in **Figure 3**b) further confirms this interpretation. The incident Gaussian beam is diffracted efficiently into the grating region and then transferred into the guided mode with relatively weak parasitic standing-wave features. The field remains spatially smooth along the coupling direction, which explains why the 2D spectrum is also smooth. At this stage, the device appears well optimized; however, this result still reflects an effectively infinite-width structure and therefore does not guarantee equivalent behavior in a practical 3D implementation.

3.2. Performance degradation in the 3D straight-grating coupler

When the same straight-grating design is evaluated in 3D, its performance degrades significantly. As shown in **Figure 4**a), the peak transmission drops to 0.405, corresponding to about -3.92 dB, and the maximum shifts from 1550 nm to approximately 1560 nm. The 3 dB bandwidth is 54.90 nm, which is similar in width to the 2D result, but the spectral profile is no longer smooth: pronounced oscillations appear across the passband. This comparison shows that the main issue is not simply

bandwidth narrowing, but rather the loss of spectral stability and peak efficiency after realistic 3D effects are introduced.

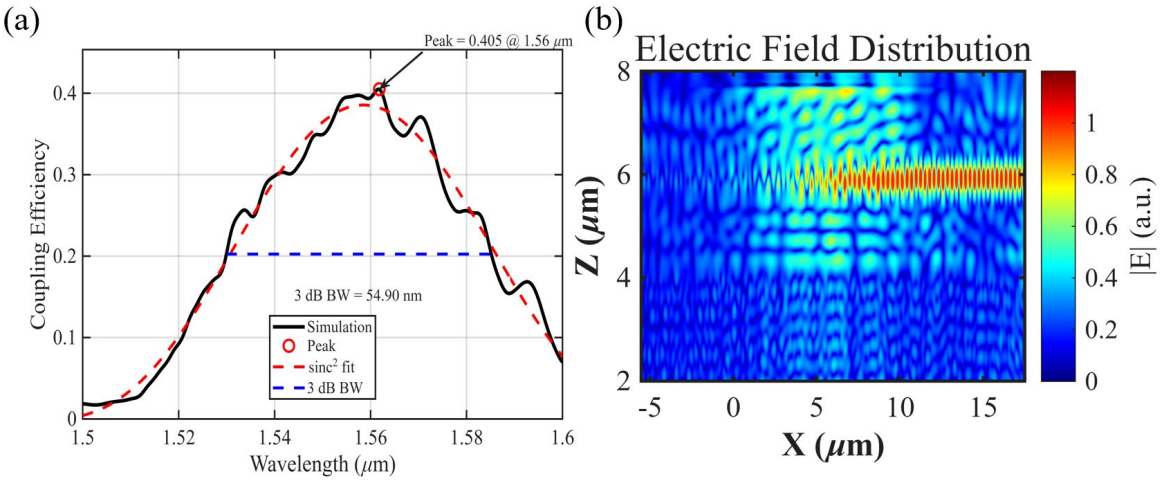

Figure 4. 3D straight-grating coupler. **a**) Simulated transmission spectrum of the 3D straight grating, showing a reduced peak transmission of 0.405 and a red-shifted maximum near 1.56 μm. **b)** Electric-field distribution of the same device, showing strong superposed interference fringes consistent with Fabry - Perot parasitic reflections across the whole space.

The field map in **Figure 4**.b) explains the reason for this degradation. Compared with the smooth 2D field in **Figure 3**.b), the 3D straight grating exhibits chaotic interference fringes in the surrounding area. These features indicate strong parasitic interference and are consistent with Fabry–Perot oscillations caused by reflection and multi-path re-interference, a behavior that is well known in grating couplers when back-reflection is not sufficiently suppressed. In practical terms, this means that a straight grating that appears satisfactory in 2D may still be unsatisfactory for fabrication if 3D validation is omitted.

3.3. Recovery by the 3D focusing-apodized grating coupler

The optimized 3D focusing-apodized grating substantially restores the desired device behavior. As shown in **Figure 5**.a), the peak transmission increases to 0.555 at 1550 nm (corresponding to −2.56 dB), while the 3 dB bandwidth broadens to 61.05 nm. Importantly, the spectral maximum returns precisely to the target wavelength, and the

oscillatory modulation observed in the 3D straight-grating model is strongly suppressed. The transmission spectrum again approaches a smooth sinc^2 envelope, indicating that the grating has recovered a more controlled coupling profile under full 3D conditions.

The peak efficiency remains below the 0.69 value achieved in the 2D model because the focusing grating includes a wedge-shaped taper. After focusing, the light must be compressed into a narrow 500 nm wide single-mode straight waveguide. This process introduces a pronounced lateral mode mismatch before entering the confined waveguide mode, resulting in additional scattering loss typically on the order of 15-20 %.

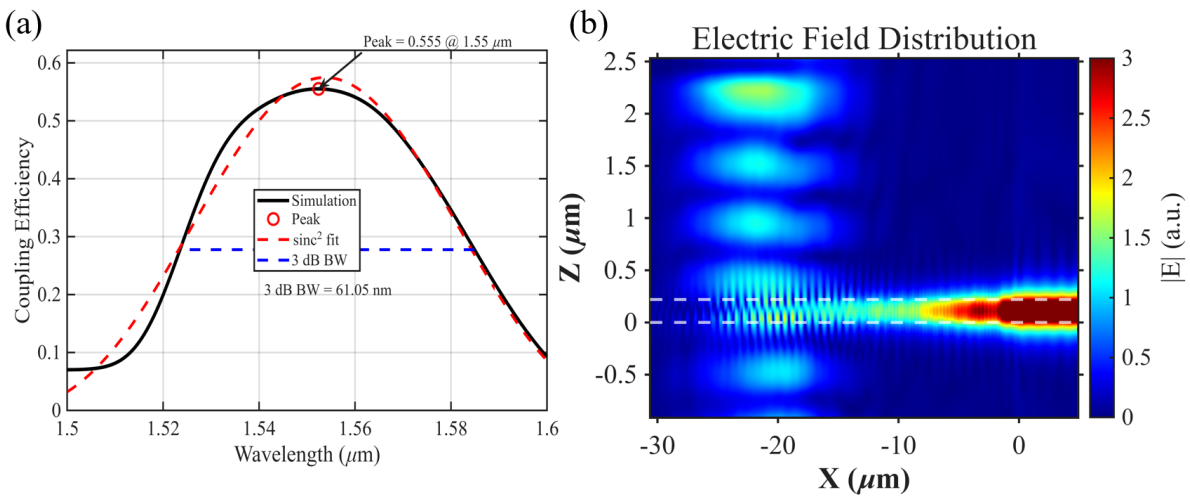

Figure 5. Simulation result of optimized 3D focusing-apodized grating coupler. **a**) Simulated transmission spectrum of the optimized 3D focusing-apodized grating, showing a peak transmission of 0.555 at 1550 nm and a 3 dB bandwidth of 61.05 nm. **b**) Electric-field distribution of the optimized structure, showing clean focusing into the taper entrance and reduced parasitic interference.

The field profile in **Figure 5**b) shows why this redesign is effective. The incident beam is first redistributed by the curved grating lines and then focused toward the taper entrance, after which the optical power is compressed into the single-mode output waveguide. Relative to **Figure 4**b), the interference pattern is much cleaner and the radiated field is more spatially organized. This indicates that the redesign improves mode matching, reduces parasitic reflections. The result is therefore not merely a higher

peak transmission, but a more fabrication-relevant 3D optical structure with improved wavelength targeting and reduced spectral ripple. The role of the focusing geometry this work design is thus confirmed by the full-wave simulation in **Figure 5**.

4. Discussion

4.1. Comparison with representative high-efficiency grating couplers

Table 1 benchmarks the present design against representative high-efficiency grating couplers reported in the literature. A clear trend is that the highest efficiencies are typically achieved through engineered vertical asymmetry or reflection enhancement, including poly-silicon overlays, metal back-reflectors, DBR stacks, dual-etch processes, or slanted gratings.[2] While effective in improving upward directionality and suppressing substrate leakage, these approaches inevitably introduce additional fabrication complexity, tighter alignment tolerances, and dependence on non-standard process modules.

In contrast, the present simulation-based design is strictly confined to a single shallow-etched SOI platform, without invoking reflectors, overlays, DBRs, or multi-step etching. Under these constraints, it achieves a peak coupling efficiency of 0.555 at 1550 nm, together with a 3 dB bandwidth of 61.05 nm, which is competitive compared with other models in **Table 1**. More importantly, the simulated spectral response is markedly smoother than that of the 3D straight coupler (**Figure 4**.a), indicating that the proposed strategy improves not only efficiency but also spectral quality. These results suggest that, even without complex enhancements, a favorable balance between efficiency, bandwidth, and spectral smoothness is achievable, supporting the practical feasibility of the design within standard fabrication platforms.

Table 1. Comparison with representative high-efficiency grating couplers.[2]

Scheme	Additional process / special structure	Peak CE (dB)	BW (nm)	Main characteristics	Refs.

SOI GC with poly-Si overlay	Overlay deposition	Sim.[1] - 1.0 Exp.[1] - 1.6	44 (1 dB)	Higher upward directionality, but requires extra overlay process	[3]
SOI GC with Au bottom reflector	Metal reflector / bonding	Sim. - 1.1 Exp. - 1.6	45 (1 dB)	Very high directionality, but fabrication is more complex	[4]
SiN GC with DBR reflector	Multilayer DBR	Sim. - 2.3 Exp. - 2.5	53 (1 dB)	Reflector-assisted, multilayer deposition required	[5]
SiN GC with SOI grating reflector	SOI reflector	Sim. - 2.3 Exp. - 1.3	80 (1dB)	Reflector-assisted	[6]
SOI GC, duty-cycle apodized	Apodization only	Exp. - 3.1	41 (1 dB)	Better mode overlap without reflector	[7]
SOI SWGC, effective-index apodized	SWG + apodization	Sim. - 2.0 Exp. - 2.2	64 (3 dB)	Effective-index engineering improves overlap and bandwidth	[8]
SOI SWGC,	Multiple etch depths	Sim. - 1.1	52 (3 dB)	High CE, but needs more complex etch	[9]

dual-etch		Exp. - 1.3		process	
Slanted GC	Slanted / FIB-related structure	Sim. - 1.9 Exp. - 3.4	80 (3 dB)	High directionality, but fabrication is difficult	[10]
This work	**No reflector; no overlay; no DBR; no dual-etch; focusing + apodized shallow-etched SOI GC**	**Sim. - 2.56 (T ~0.555)**	**61.05 (3 dB)**	**Peak locked at 1550 nm; smooth sinc^2 spectrum; Fabry-Perot oscillation strongly suppressed**	**/**

[1] Sim. refers to simulation results, Exp. refers to experimental results.

4.2. Physical insights and tape-out implications

The comparison above indicates that the performance limitations commonly observed in single-etch SOI grating couplers largely comes from uncontrolled 3D parasitic interference and lateral mode mismatch. As revealed by the failure of direct 2D-to-3D translation, the assumption of infinite extent in the transverse (x–y) plane neglects multi-path reflections at the device boundaries, leading to Fabry–Pérot-like resonances and pronounced spectral oscillations.

These effects are mitigated through the combined use of focusing and apodization. The focusing geometry reshapes the diffracted wavefront, suppressing divergence and reducing leakage, thereby improving mode overlap with the fiber. Meanwhile, the chirped (apodized) fill factor gradually modifies the effective index and scattering strength along the propagation direction, effectively breaking the parasitic effects and suppressing coherent back-reflections.

As a result, the simulation yields a smooth, oscillation-free spectrum with a peak

coupling efficiency of 0.555 precisely at 1550 nm and a 61.05 nm bandwidth. From a practical perspective, such spectral characteristics are highly desirable for wafer-level testing and WDM system integration, where suppressing spectral ripples and wavelength shifts are needed. Importantly, these results are obtained within a standard 220-nm SOI platform with a single 80-nm shallow etch, indicating that the proposed approach is fully compatible with conventional fabrication flows. Therefore, this study demonstrates a physically grounded and fabrication-available pathway to achieving robust and spectrally stable grating couplers without relying on complex modifications.

5. Conclusion

In summary, we have presented a systematic methodology to bridge the gap between idealized 2D designs and realistic 3D physical behaviors in silicon grating couplers. By identifying and mitigating the severe spectral degradation caused by non-ideal diffraction and parasitic effects, we designed a high-efficiency 3D focusing-apodized grating coupler fully compatible with standard 220-nm SOI foundries. Without relying on exotic reflectors or complex multi-etch processes, the optimized device achieves a simulated peak transmission of 0.555 (–2.56 dB) accurately targeted at 1550 nm, alongside a broad 3-dB bandwidth of 61.05 nm and a remarkably smooth spectral envelope. This approach demonstrates that wavefront engineering via focusing and apodization may unlock high-performance, spectrally stable optical I/O interfaces, offering a possibly reliable and tape-out-ready solution for silicon photonic integrated circuits.

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